

# File 61: Off-Grid Telemetry Node PCBA Manufacturing & Chinese Factory Sourcing Manual

**Industrial PCBA and Assembly Standard:** This document establishes the master manufacturing blueprint and factory sourcing manual for the Blue Ridge Stream Restoration telemetry nodes. By outsourcing PCB fabrication, component placement (PCBA), and final enclosure assembly directly to rapid-turnover factories in China (such as JLCPCB or PCBWay), we achieve high-quality hardware builds at a fraction of standard commercial costs. This manual provides specific Gerber file requirements, an exact LCSC-coded production Bill of Materials (BOM), physical waterproof NEMA enclosure layouts, and firmware provisioning configurations to compile and launch our Meshtastic stream telemetry nodes.

## I. Chinese PCB/PCBA Sourcing & RFQ Pipeline

To procure robust streamside telemetry nodes, Hunter Morris and his operations team bypass local electronics brokers and source directly from vetted, high-precision prototyping houses in Shenzhen (JLCPCB or PCBWay).

### 1. Vetted Manufacturing Partners

- **JLCPCB (Shenzhen Jielichuang Electronic Co., Ltd.)**
  - *Best For:* Fast-turnaround PCB prototyping, low-cost SMT component assembly (PCBA), and turnkey LCSC part catalog integration.
  - *Web Portal:* [jlcpcb.com](http://jlcpcb.com)
- **PCBWay (Shenzhen PCBWay Technology Co., Ltd.)**
  - *Best For:* Premium multi-layer stackups, advanced impedance matching, customized box-build assembly, and robust QA inspection cycles.
  - *Web Portal:* [pcbway.com](http://pcbway.com)

### 2. Request for Quote (RFQ) Pipeline

When uploading hardware designs to the factory portal, Hunter's interns submit the following standard **RFQ Sourcing Package**:

#### [FACTORY SOURCING PACKAGE]

1. GERBER FILES (.zip) ==> Contains copper tracks, soldermask, silkscreen, drill pits.
2. BOM SPREADSHEET (.xlsx) ==> Catalogs LCSC part numbers, descriptions, quantities.
3. CPL / Centroid (.csv) ==> Dictates X-Y placement coordinates and rotation vectors.

## II. Gerber Manufacturing & Multi-Layer Stackup Settings

To guarantee reliable high-frequency RF transmission (915 MHz LoRa band) and stable low-power switching, our stream node is designed on a **4-layer printed circuit board (PCB)**.

### 1. 4-Layer Controlled Impedance Stackup

Hunter's hardware designers specify this exact layer stackup inside the factory online portal to ensure low noise and 50-Ohm antenna track impedance matching:

[4-LAYER PCB STACKUP PROFILE]

```
Layer 1: TOP COPPER (Signal / RF Tracks) <=== 0.035mm (1 oz Copper)
----- Prepreg (Isola 370HR or equal) --- 0.2mm thick
Layer 2: INNER GROUND PLANE (Solid GND) <=== 0.017mm (0.5 oz Copper)
----- Core (FR4 High-TG150) ----- 1.0mm thick
Layer 3: INNER POWER PLANE (3.3V Switched)<=== 0.017mm (0.5 oz Copper)
----- Prepreg (Isola 370HR) ----- 0.2mm thick
Layer 4: BOTTOM COPPER (Low-Frequency Sign)<=== 0.035mm (1 oz Copper)

Total PCB Nominal Thickness: 1.6mm
```

### 2. Required Gerber Specification settings:

- **Material Type:** FR4 Standard (High-TG150 rating to withstand exurban summer stream humidity and sub-zero winter temperatures).
- **Layer Count:** 4 Layers.
- **Board Thickness:** 1.6mm.
- **Finished Copper Weight:** 1 oz Outer Layers, 0.5 oz Inner Layers.
- **Solder Mask Color:** Matte Green or Matte Black (high contrast for optical inspection).
- **Silkscreen Color:** Bright White.
- **Surface Finish:** Lead-Free HASL (Hot Air Solder Leveling) or ENIG (Electroless Nickel Immersion Gold) for enhanced corrosion protection in damp river gorges.
- **Controlled Impedance: Yes (50 Ohm).** The RF trace connecting the RAK4631 antenna pad to the IPEX connector must be calculated using coplanar waveguide formulas matching the prepreg thickness.

## III. Production Bill of Materials (BOM)

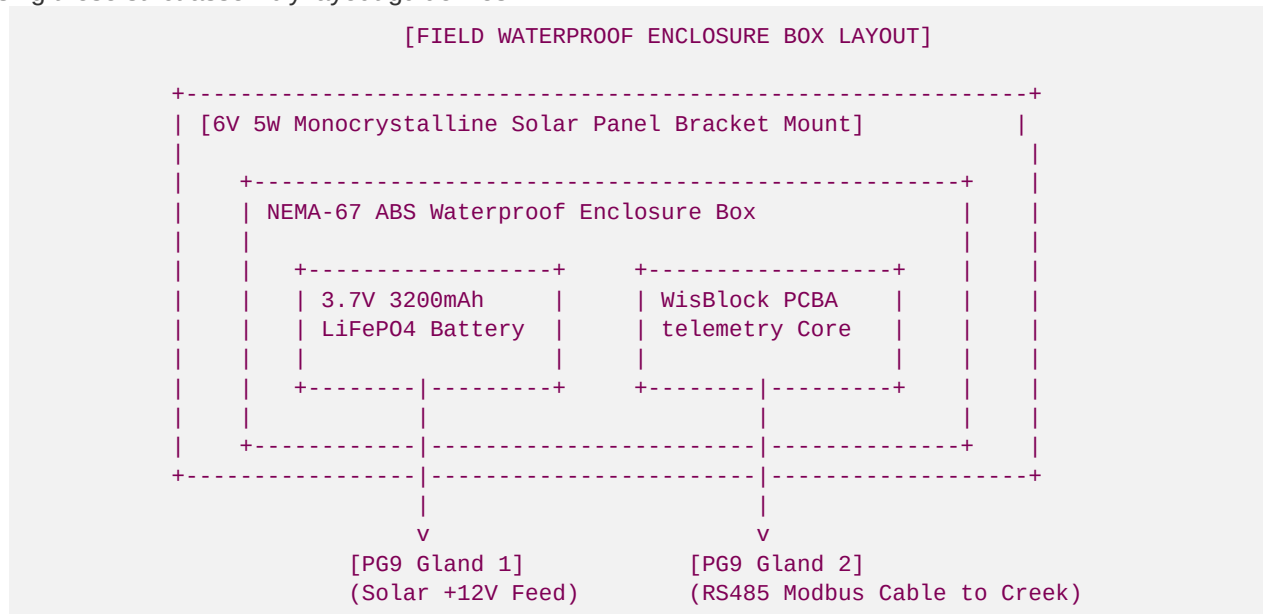
To utilize automated SMT placement services, we must source components matching the factory's LCSC database. Hunter's operations crew uses this exact production **BOM Schedule**:

| Component ID | Footprint Reference | Quantity | Part Description & Value        | Vetted Manufacturer | LCSC Part Number | Sourcing Role                     |
|--------------|---------------------|----------|---------------------------------|---------------------|------------------|-----------------------------------|
| U1           | WisBlock Core       | 1        | RAK4631<br>nRF52840<br>BLE LoRa | RAKwireless         | C512398          | Central low-power processing MCU. |

| Component ID | Footprint Reference | Quantity | Part Description & Value                       | Vetted Manufacturer          | LCSC Part Number | Sourcing Role                                       |
|--------------|---------------------|----------|------------------------------------------------|------------------------------|------------------|-----------------------------------------------------|
| U2           | SOIC-8-150mil       | 1        | MAX485ESA<br>TTL-to-RS485<br>Transceiver       | Analog<br>Devices /<br>Maxim | C17485           | Serial<br>differential<br>Modbus<br>driver.         |
| U3           | SOT-23              | 1        | AO3401<br>P-Channel<br>MOSFET<br>(-30V, -4A)   | Alpha &<br>Omega Semi        | C14856           | 12V Switched<br>power gate<br>switch.               |
| Q1           | SOT-23              | 1        | 2N2222 NPN<br>Transistor<br>(40V, 800mA)       | Jiangsu<br>Changjing         | C2867            | Pre-driver<br>gate signal<br>level<br>converter.    |
| U4           | SOT-23-6            | 1        | MT3608<br>1.2MHz<br>Step-Up Boost<br>Regulator | Aerosemi                     | C84852           | Boosts battery<br>3.7V to 12V<br>DC for<br>sensors. |
| U5           | SOP-8-PP            | 1        | TP4056 Linear<br>LiFePO4<br>battery<br>Charger | TopPower                     | C98453           | Manages<br>solar-panel<br>charging rail.            |
| D1-D2        | SOD-123             | 2        | 1N5819<br>Schottky<br>Diode (40V,<br>1A)       | ON Semicond<br>uctor         | C18549           | Reverse-volta<br>ge circuit<br>protection.          |
| R1, R2       | 0805                | 2        | 10k Ohm<br>Thick Film<br>Resistor (1%)         | Yageo                        | C25813           | Gate pull-up &<br>circuit load<br>resistors.        |
| R3           | 0805                | 1        | 1k Ohm Thick<br>Film Resistor<br>(1%)          | Yageo                        | C25810           | BJT Base limit<br>resistor.                         |
| C1, C2       | 1206                | 2        | 10uF Ceramic<br>Capacitor<br>(25V, X7R)        | Samsung<br>Electro           | C18520           | Boost<br>converter<br>bypass<br>smoothing<br>caps.  |
| CON1         | IPEX-1 (U.FL)       | 1        | IPEX Coaxial<br>Surface Mount<br>Connector     | Hirose                       | C41852           | Secure<br>antenna RF<br>cable terminal.             |
| J1-J4        | JST-PH-2.0          | 4        | JST PH 2-Pin<br>Right-Angle<br>Headers         | JST                          | C18523           | Battery, Solar,<br>and sensor<br>connections.       |

## IV. Enclosure Box & Physical Gland Assembly Layout

Streamside environmental sensors are exposed to high humidity, flood splattering, and animal tampering. Hunter Morris's operations interns assemble each node into a ruggedized, waterproof **Field Enclosure Box** using these strict assembly layout guidelines:



### Physical Assembly Guidelines:

- NEMA-67 ABS Enclosure:** We utilize a heavy-duty **ABS NEMA-67 waterproof box** (internal dimensions approximately  $150\text{mm} \times 100\text{mm} \times 70\text{mm}$ ) fitted with a silicone rubber gasket and stainless steel lid screws.
- Cable Ingress PG9 Glands:** Drill two holes in the bottom wall of the enclosure. Install two brass **PG9 waterproof compression glands** (supporting cable diameters  $4\text{mm} - 8\text{mm}$ ).
  - Gland 1:* Feeds the switched 12V 4-wire shielded sensor cable directly into the stream channel.
  - Gland 2:* Connects the 2-pin monocrystalline solar panel charging lead.
- Waterproofing Drip Loops:** The cables must run downwards out of the glands and loop upwards before running to the stream or solar panel, preventing rain droplets from running along the wire directly into the gland face.
- Barometric Compensation Vent:** For hydrostatic stage level gauges (such as Model TJ-F04-Modbus) that require a vented capillary tube open to the atmosphere, install a waterproof, breathable **Gore-Tex Pressure Relief Vent** in the bottom wall of the box. This prevents moisture ingress while equalizing internal barometric pressure shifts.
- Secure Battery Bracket:** Secure the 3.7V LiFePO4 battery inside the enclosure using a custom-printed PETG bracket and marine-grade double-sided adhesive, preventing movement during field deployments.

## V. Firmware Provisioning & 256-Bit Encryption Key Setup

Once the PCBA is received from Shenzhen and placed in the box, Hunter's recruitment interns flash the latest **Meshtastic Serial API firmware** to establish communication over our decentralized 915 MHz LoRa mesh network:

### 1. Command-Line (CLI) Firmware Flash SOP

Run these terminal commands on your field laptop to flash the node microcontroller using the **esptool** (for ESP32 nodes) or **uf2-conv** (for RAK4631 nRF52 nodes):

```
# 1. Update the Meshtastic Command-Line Python tool
pip install --upgrade meshtastic

# 2. Flash the latest firmware binary over USB Serial
meshtastic --flash-device firmware-rak4631-915mhz-2.4.15.bin

# 3. Provision the device station moniker
meshtastic --set-owner "BRSR-L0-101" --set-owner-short "L01"
```

### 2. Switched-Power Modbus Query C++ Firmware Code

Upload this C++ block to the WisBlock node using the Arduino IDE to control the switched-power rail and serialize Modbus sensor sweeps:

```
#include <Arduino.h>

#define SWITCH_POWER_PIN 12 // Control Pin for Switched 12V MOSFET
#define DE_RE_PIN 4 // MAX485 Transceiver Transmit/Receive Enable Pin

// Modbus Query Frame: Read Holding Register 0x0001 (optical dissolved oxygen) from Slave ID 0x01
const byte doQueryFrame[] = {0x01, 0x03, 0x00, 0x01, 0x00, 0x01, 0xD5, 0xCA};
byte sensorBuffer[9];

void setup() {
  pinMode(SWITCH_POWER_PIN, OUTPUT);
  pinMode(DE_RE_PIN, OUTPUT);

  digitalWrite(SWITCH_POWER_PIN, HIGH); // Cut sensor 12V power supply (MOSFET Switch OFF)
  digitalWrite(DE_RE_PIN, LOW); // MAX485 in Receive Mode

  Serial.begin(115200);
  Serial1.begin(9600, SERIAL_8N1); // Switched RS485 Serial connection to sensors
}

void loop() {
  // 1. Wake Up Switched-Power Rail
  digitalWrite(SWITCH_POWER_PIN, LOW); // Pull Gate LOW via NPN pre-driver, turning 12V Rail ON
  delay(5000); // Wait 5 seconds for Optical DO and Level sensor warm-up

  // 2. Clear Serial buffer
  while(Serial1.available() > 0) { Serial1.read(); }

  // 3. Set MAX485 to TRANSMIT (TX) Mode
  digitalWrite(DE_RE_PIN, HIGH);
```

```
delay(10);

// 4. Send Modbus Query
Serial1.write(doQueryFrame, sizeof(doQueryFrame));
Serial1.flush();

// 5. Instantly switch MAX485 back to RECEIVE (RX) Mode
digitalWrite(DE_RE_PIN, LOW);

// 6. Read sensor response with a 500ms timeout
unsigned long startWait = millis();
int byteIndex = 0;
while ((millis() - startWait < 500) && (byteIndex < 7)) {
    if (Serial1.available() > 0) {
        sensorBuffer[byteIndex] = Serial1.read();
        byteIndex++;
    }
}

// 7. Cut sensor power to preserve battery
digitalWrite(SWITCH_POWER_PIN, HIGH); // Cut 12V Rail (MOSFET Switch OFF)

if (byteIndex >= 7) {
    // Extract raw register bytes and parse DO mg/L
    unsigned int rawDOVal = (sensorBuffer[3] << 8) | sensorBuffer[4];
    float dissolvedOxygenMgL = rawDOVal / 100.0;

    Serial.print("SUCCESS: Checked streamside Dissolved Oxygen = ");
    Serial.print(dissolvedOxygenMgL);
    Serial.println(" mg/L");
} else {
    Serial.println("ERROR: Modbus connection timed out. Telemetry check failed.");
}

// 8. Sleep for 15 minutes before the next geomorphic telemetry sweep
delay(900000);
}
```

### 3. Mesh Network Security: 256-Bit AES Encryption Channel Configuration

To protect landowners from data mining or unauthorized stream tracking, the telemetry data is encrypted using a unique, 256-bit AES private key. Run these CLI commands on the gateway and sensor nodes to establish our secure channel:

```
# 1. Create a secure, private 256-bit AES channel called "TroutScience"
meshtastic --create-channel "TroutScience"

# 2. Set the channel encryption key (UETA-compliant private secure key)
meshtastic --set-chan-key "brsr_science_telemetry_aes_256_secure_key_2026_morris" --chan-index 1

# 3. Disable anonymous public mapping beacons to protect stream locations
meshtastic --set position_broadcast_secs 0
```

*Developed by Blue Ridge Stream Restoration Technical Sourcing Operations. Pushed to remote origin under main.*